

Week_2_Data_Ingestion

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1 Market Sentiment Predictor

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The goal of this project is to predict the directional price movement of a specific stock using both historical price data and news information. Two separate models will predict price movements based on historical data and analyze news sentiment.

```
[33]: import pandas as pd
import os
```

```
[34]: os.getcwd()
```

```
[34]: '/home/jupyter-previt/market-sentiment-predictor/notebooks'
```

2 Week 2 - Data Ingestion

There are two main datasets to this project. The first is a dataset containing news article content including titles, content, and sentiment scores from various news sources around the AAPL stock. The second dataset (introduced later in the notebook) contains a time series of AAPL stock prices. The final dataset will combine the two datasets on date.

2.1 Sentiment Dataset

```
[2]: #Github didn't allow us to upload a file more than 25MB. This is combining all
    ↪files into one file
raw_data_1 = pd.read_csv('../data/apple_news_data_1.csv', encoding='latin-1')
raw_data_2 = pd.read_csv('../data/apple_news_data_2.csv', encoding='latin-1')
raw_data_3 = pd.read_csv('../data/apple_news_data_3.csv', encoding='latin-1')
raw_data_4 = pd.read_csv('../data/apple_news_data_4.csv', encoding='latin-1')
raw_data_5 = pd.read_csv('../data/apple_news_data_5.csv', encoding='latin-1')

raw_data = pd.concat([raw_data_1, raw_data_2, raw_data_3, raw_data_4,
    ↪raw_data_5])[['date', 'title', 'content', 'link', 'symbols', 'tags',
    ↪'sentiment_polarity', 'sentiment_neg', 'sentiment_neu', 'sentiment_pos']]

[3]: #view a small subsection of the data to ensure formatting looks correct
raw_data.head(10)
```

[3]:

	date \		title \		content \		link \
0	2021-04-30T21:38:00+00:00		Techâ s Big Five Had Fantastic Pandemics. Her...	0	Alphabet, Amazon.com, Apple, Facebook, and Mic...		https://finance.yahoo.com/m/bbe3f7c9-c7af-3c9f...
1	2021-04-30T20:41:52+00:00		Valuation Remains a Concern for Churchill Capi...	1	Churchill Capital IV (NYSE:CCIV) stock has gra...		https://finance.yahoo.com/news/valuation-remai...
2	2021-04-30T20:18:00+00:00		Big techâ s trillion-dollar pandemic year may...	2	This week concludes the first quarter earnings...		https://finance.yahoo.com/m/6e441d74-37c1-3642...
3	2021-04-30T19:32:30+00:00		What Appleâ s blowout earnings means for Berk...	3	Krish Sankar, Managing Dir. & Cowen Senior Res...		https://finance.yahoo.com/video/apple-blowout-...
4	2021-04-30T18:49:00+00:00		EU Agrees With Spotify: Apple Is Abusing Its A...	4	The EU's European Commission just released pre...		https://finance.yahoo.com/m/07524444-63a7-34aa...
5	2021-04-30T18:30:06+00:00		ETFs to Stack as Reopening Poses No Hindrance ...	5	The tech-heavyÂ Nasdaq Composite Index was a s...		https://finance.yahoo.com/news/etfs-stack-reop...
6	2021-04-30T18:19:10+00:00		Big Tech Stocks Flash Red Flag to Regulators A...	6	If â Big Techâ was a gamefish, like Faceboo...		https://finance.yahoo.com/news/big-tech-stocks...
7	2021-04-30T18:13:06+00:00		Top Stock Reports for Apple, Alphabet & Bl...	7	Friday, April 30, 2021\n\nThe Zacks Research D...		https://finance.yahoo.com/news/top-stock-repor...
8	2021-04-30T17:54:31+00:00		Why Stem Stock Looks Very Attractive Following...	8	Not infrequently, Wall Street focuses excessiv...		https://finance.yahoo.com/news/why-stem-stock-...
9	2021-04-30T17:34:59+00:00		â Technology is the largest single growth tre...	9	Keith Fitz-Gerald, Fitz-Gerald Group Chief Inv...		https://finance.yahoo.com/video/technology-lar...

```

                                symbols \
0      AAPL.US, AAPL34.SA, AMZN.US, FB.US, GOOGL.US
1      AAPL.US, CCIV-UN.US, CCIV.US
2      AAPL.US, AAPL34.SA, AMZN.US, MSFT.US
3      AAPL.US, AAPL34.SA, BRK-A.US, BRK-B.US, DJI.IN...
4      AAPL.US, AAPL34.SA, SPOT.US
5      AAPL.US, AAPL34.SA, COMP.US, FB.US, GOOGL.US, ...
6      AAPL.US, CRM.US, FB.US, GME.US, GOOGL.US, MSFT...
7      AAPL.US, AAPL34.SA, BLK.US, F.US, GOOGL.US, ZT...
8      AAPL.US, AMZN.US, BX.US, FSLR.US, GE.US, GEOO3...
9      AAPL.US, AAPL34.SA, AMZN.US, DJI.INDX, FB.US, ...

```

```

                                tags sentiment_polarity \
0      AMAZON, APPLE, FACEBOOK, HARLEY FINKELSTEIN, M...      0.395
1      BUSINESS COMBINATION, CCIV, CHURCHILL CAPITAL ...      0.998
2      AMAZON, APPLE INC, BIG TECH, MARKET CAPITALIZA...      0
3      ALEXIS CHRISTOFOROUS, BERKSHIRE HATHAWAY, COWE...      0
4      APP STORE, APPLE, EUROPEAN COMMISSION, MUSIC S...     -0.71
5      APPLE, CONSENSUS ESTIMATE, EARNINGS PER SHARE,...      0.999
6      BITCOIN, DIGITAL TRANSFORMATION, FACEBOOK, MIC...      0.999
7      ALPHABET, APPLE, APPLE WATCH, BLACKROCK, RESEA...      0.999
8      CLEAN-ENERGY, STEM STOCK      0.999
9      ALEXIS CHRISTOFOROUS, KEITH FITZ-GERALD, SIBIL...      0

```

```

sentiment_neg sentiment_neu sentiment_pos
0      0      0.871      0.129
1      0.023      0.85      0.127
2      0      1      0
3      0      1      0
4      0.093      0.885      0.022
5      0.027      0.826      0.148
6      0.026      0.861      0.113
7      0.042      0.775      0.183
8      0.015      0.83      0.154
9      0      1      0

```

```

[4]: #check how many NA values there are. We care most about sentiment, date, and
      ↪ content columns
raw_data.isna().sum().sort_values(ascending=False)

```

```

[4]: tags      17790
      sentiment_pos      5853
      sentiment_neu      5812
      sentiment_neg      5767
      sentiment_polarity      5721
      symbols      5539

```

```

link                5340
content             4901
title              4255
date               2808
dtype: int64

```

```

[5]: #force sentiment scores to be a numerical value, NA if not
for col in
    ↳ ['sentiment_pos', 'sentiment_neu', 'sentiment_neg', 'sentiment_polarity']:
        raw_data[col] = pd.to_numeric(raw_data[col], errors='coerce')

#convert dates to a datetime object and content to a string
raw_data['date'] = pd.to_datetime(raw_data['date'], errors='coerce', utc=True).dt.
    ↳ date
raw_data['content'] = raw_data['content'].astype('str')

```

```

[27]: #this data is messy, drop observations that have NAs in critical columns
raw_data = raw_data.
    ↳ dropna(subset=['sentiment_pos', 'sentiment_neu', 'sentiment_neg', 'sentiment_polarity', 'content'])

#print out the data type of each column to ensure conversions from above worked
for col in raw_data.columns:
    print(f'Column: {col} , Data Type: {raw_data[col].dtype}')

```

```

Column: date , Data Type: object
Column: title , Data Type: object
Column: content , Data Type: object
Column: link , Data Type: object
Column: symbols , Data Type: object
Column: tags , Data Type: object
Column: sentiment_polarity , Data Type: float64
Column: sentiment_neg , Data Type: float64
Column: sentiment_neu , Data Type: float64
Column: sentiment_pos , Data Type: float64
Column: Net Sentiment , Data Type: float64

```

```

[7]: #Create a single sentiment score that we can use to determine if an article
    ↳ contains more negative or positive sentiment
raw_data['Net Sentiment'] = raw_data['sentiment_pos'] -
    ↳ raw_data['sentiment_neg']

#while not the main target variable, net sentiment could be interesting to
    ↳ determine presence of positive and negative language
#we can also make this a classification model by converting to 'Positive',
    ↳ 'Neutral', or 'Negative' here.

```

The “content” column will be the predictor variable in the first model, and the string will be transformed into a numerical matrix using feature engineering methods such as TF-IDF. The “snetiment polarity” variable will be the target variable. The goal of the first model will be to predict the

sentiment polarity of a news article given its content.

```
[8]: print(raw_data.shape)
```

```
(29700, 11)
```

This dataset has 29,700 rows and 10 columns (plus the net sentiment score column created by us)

2.2 Price Timeseries Dataset

```
[9]: ts_data = pd.read_csv('../data/AAPL Stock Data.csv')
```

```
[10]: ts_data.head(10)
```

```
[10]:
```

	Dates	AAPL Share Price
0	Jan-03-2005	1.13
1	Jan-04-2005	1.14
2	Jan-05-2005	1.15
3	Jan-06-2005	1.15
4	Jan-07-2005	1.24
5	Jan-10-2005	1.23
6	Jan-11-2005	1.15
7	Jan-12-2005	1.17
8	Jan-13-2005	1.25
9	Jan-14-2005	1.25

```
[11]: #check how many NA values there are
ts_data.isna().sum().sort_values(ascending=False)
```

```
[11]: Dates          0
AAPL Share Price    0
dtype: int64
```

```
[15]: #Convert values to the expected formatting
ts_data['Dates'] = pd.to_datetime(ts_data['Dates']).dt.date
ts_data['AAPL Share Price'] = ts_data['AAPL Share Price'].astype('float')
```

```
[22]: for col in ts_data.columns:
      print(f'Column: {col} , Data Type: {type(ts_data[col][0])}')

```

```
Column: Dates , Data Type: <class 'datetime.date'>
```

```
Column: AAPL Share Price , Data Type: <class 'numpy.float64'>
```

```
[31]: print(ts_data.shape)
```

```
(5298, 2)
```

There are 5,298 rows in this dataset and two columns (date and price).

2.3 Combining Datasets

```
[19]: #Get the average sentiment by day, makes it possible to join on AAPL daily stock prices
sentiment_data = raw_data.groupby(['date'])[['sentiment_polarity']].mean()
print(sentiment_data.shape)
```

(1574, 1)

```
[17]: #join on date
all_data = pd.merge(sentiment_data, ts_data, how = 'right', left_on = 'date', right_on = 'Dates')
```

```
[30]: for col in all_data.columns:
      print(f'Column: {col} , Data Type: {type(all_data[col][0])}')
```

Column: sentiment_polarity , Data Type: <class 'numpy.float64'>

Column: Dates , Data Type: <class 'datetime.date'>

Column: AAPL Share Price , Data Type: <class 'numpy.float64'>

```
[18]: all_data
```

```
[18]:
```

	sentiment_polarity	Dates	AAPL Share Price
0	NaN	2005-01-03	1.13
1	NaN	2005-01-04	1.14
2	NaN	2005-01-05	1.15
3	NaN	2005-01-06	1.15
4	NaN	2005-01-07	1.24
...	...	...	...
5293	NaN	2026-01-16	255.53
5294	NaN	2026-01-20	246.70
5295	NaN	2026-01-21	247.65
5296	NaN	2026-01-22	248.35
5297	NaN	2026-01-23	247.98

[5298 rows x 3 columns]

We have a greater amount of stock price data than news sentiment data on the AAPL stock. There are currently 5,298 rows and 3 columns, but only 1,574 rows have sentiment data (as shown from the print statement above). We can remove dates without any sentiment analysis at a future date. For right now, this data could be helpful for a traditional forecasting model

```
[35]: all_data['sentiment_polarity'].describe()
```

```
[35]: count    1133.000000
      mean      0.528144
      std      0.301737
      min     -0.993000
      25%      0.409429
```

```
50%          0.550500
75%          0.685174
max           1.000000
Name: sentiment_polarity, dtype: float64
```

The sentiment polarity looks to be performing as expected. Scores range from -.99 to 1.00, but most are positive (which is expected due to Apple's stellar performance during this time period).

All models will use next-day stock price direction as the main target variable. There will be three possible forecasting models, the traditional model will exclusively use previous stock prices as predictor variables, the second model exclusively uses sentiment analysis as the predictor variable, while the third model will combine both.

```
[ ]:
```